



## AI Playbook and Toolkit for Public Health Departments

# Privacy-Preserving AI and De-identification

GOV 310

Privacy-preserving AI is the set of technical, policy, and workflow practices that allow public health agencies to use data responsibly while reducing the risk that individuals, households, providers, or communities can be identified or harmed. This module focuses on how privacy protections must be built into AI planning before data are extracted, shared, used for model development, connected to a vendor platform, or summarized through a generative AI tool. Public health agencies often have legal authority to collect and use sensitive data, but authority to use data does not remove the obligation to protect privacy, confidentiality, public trust, and community relationships. AI can increase privacy risk because it may combine data sources, infer sensitive attributes, generate outputs that reveal source details, or make sensitive information easier to retrieve. Learners in this module examine how de-identification, access control, data minimization, aggregation, suppression, secure environments, and differential privacy can support safer AI use. This module is intended for informaticians, epidemiologists, technologists, privacy officers, data stewards, and AI governance participants. It does not replace legal review. It gives technical and program staff enough understanding to recognize privacy risk, ask better implementation questions, and design safeguards that can be reviewed by privacy, legal, security, and governance leaders.

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|--------------------------|-------------------------------|
| <b>Level</b>             | intermediate/practitioner     |
| <b>Primary track</b>     | Governance and Security Track |
| <b>Appears in tracks</b> | operations-data-quality       |

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## Learning Objectives

- Explain why AI can increase privacy and confidentiality risks in public health workflows.
- Distinguish between identifiable data, de-identified data, aggregated data, limited datasets, and synthetic data at a practical level.
- Identify technical and operational safeguards that reduce privacy risk, including data minimization, role-based access, aggregation, suppression, and secure analytic environments.
- Assess re-identification risk when public health data are linked, enriched, summarized, or shared for AI-supported use.
- Recognize when de-identification alone is not sufficient and additional governance, legal, or technical controls are needed.
- Produce a privacy-preserving AI review worksheet for a real or realistic public health workflow.

## Module Overview

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## Jurisdiction and Agency Policy Note

This module provides national-level training guidance for public health agencies and partners. It is not legal advice and does not replace review by agency legal counsel, privacy officers, procurement officials, civil rights staff, records officers, or other authorized decision-makers. Requirements may vary by state, locality, territory, Tribe, agency, funding source, data-sharing agreement, emergency authority, and organizational policy. Learners should use this module to identify the issues that require review and should confirm applicable requirements before approving, procuring, deploying, or publicly communicating about AI-supported work.

## Why This Topic Matters for Public Health AI

AI can change the privacy risk profile of a public health dataset even when the source data were already collected lawfully. A dataset that seems low risk in one context may become more sensitive when linked to geospatial information, clinical history, social vulnerability indicators, case notes, or small-area counts. Generative AI and retrieval systems can also reveal sensitive details through summaries, citations, or unexpected responses. Privacy review must therefore consider not only the dataset itself, but also how the AI system accesses, transforms, combines, stores, and exposes information.

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Privacy risk is also a public trust issue. Public health agencies rely on communities, providers, laboratories, schools, and partners to share information. If AI use is perceived as opaque, excessive, or poorly controlled, communities may become less willing to participate in programs or share sensitive information. Technical safeguards support trust by showing that the agency has intentionally limited access, reduced unnecessary exposure, and retained human accountability.

Privacy-preserving AI is not a single technique. It is a layered approach that combines data governance, data minimization, secure infrastructure, access control, audit logs, de-identification, aggregation, review of outputs, and limits on downstream use. Strong privacy protection depends on designing these controls into the workflow before implementation rather than adding them after a concern arises.

## Core Concepts

De-identification is the process of reducing the risk that data can be connected to a specific person. In public health practice, de-identification must be interpreted carefully because rare diseases, small communities, unusual dates, exact locations, and combinations of demographic fields may still make a record identifiable. A dataset can appear anonymous but still be linkable to individuals when combined with other information.

Re-identification risk is the possibility that a person, household, provider, facility, or community can be identified from data that were intended to be de-identified or aggregated. AI can increase this risk by making patterns easier to discover, by generating detailed summaries, or by combining sources that were originally separated.

Re-identification risk should be assessed before data are used for model development, vendor testing, public dashboards, or training exercises.

Data minimization means using only the data elements necessary for the approved purpose. It is a practical guardrail because many AI tools perform better when given more context, but more context also increases privacy and confidentiality risk. A public health agency should document which fields are required, which are optional, which are prohibited, and which require special approval.

Differential privacy is a statistical approach that adds carefully calibrated noise to outputs to reduce the risk that information about any one person can be inferred. It can be useful for public reporting or statistical release, but it does not solve all privacy problems. Public health staff need to understand that differential privacy involves tradeoffs between privacy protection and analytic utility.

## Public Health Example

Consider a health department that wants to use AI to summarize disease investigation notes and identify common exposure settings across cases. The notes may include symptoms, dates, employer names, schools, shelters, correctional facilities, household details, immigration concerns, or other sensitive context. Even if the AI system is internal, the agency must define who can access the notes, whether the AI output can include names or exact locations, and how summaries will be reviewed before sharing.

A privacy-preserving design might limit the AI tool to authorized case investigation staff, restrict the tool from using data outside the case management environment, require source traceability, and prevent output from including identifiers unless explicitly needed for the case workflow. Aggregated exposure summaries may require suppression rules for small counts, especially in rural areas or small communities. The goal is not to block useful analysis, but to ensure that sensitive information is used only in ways that are necessary, authorized, reviewed, and documented.

## Technical Deep Dive

Privacy-preserving AI begins with data classification. Staff should identify whether the workflow uses protected health information, personally identifiable information, confidential public health data, sensitive community data, operational data, or publicly available information. Classification should be tied to rules for storage, access, sharing,

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retention, and approved AI tools. A public AI tool may be acceptable for public guidance documents but inappropriate for case narratives or line lists.

The next step is determining the minimum necessary data for the AI task. A model that classifies complaint topics may not need names or exact addresses. A tool that summarizes case notes for an assigned investigator may need identifiers inside the secure case system, but the same identifiers should not appear in evaluation exports or training examples. Data minimization should be built into data extracts, prompts, APIs, and reports.

De-identification requires more than removing names. Dates, ZIP codes, facility names, employer names, exact locations, rare conditions, and free-text narratives can all create identification risk. Natural language processing and generative AI add additional complexity because sensitive details may be embedded in text. Agencies should consider automated redaction, structured review, suppression rules, and human spot checks before using text in AI workflows.

Privacy controls must also apply to outputs. An AI-generated summary, risk score, cluster report, or dashboard can reveal sensitive information even if the input data were controlled. Output review should consider small cells, unusual combinations of attributes, identifiable narratives, and unintended inferences. Logs should document who generated outputs, what data were used, and whether the output was shared, edited, rejected, or escalated.

## Risks, Failure Modes, and Guardrails

A major failure mode is assuming that de-identification is permanent. Data that are de-identified for one purpose may be re-identifiable in another context, especially when linked with other datasets. A guardrail is to perform use-case-specific privacy review and to document assumptions about data linkage, audience, geography, and release context.

Another failure mode is exposing sensitive data to unapproved AI tools. Staff may paste text into a convenient tool without realizing that the data include protected or confidential information. A guardrail is to provide clear approved-tool lists, data-use tiers, technical blocking where possible, and training on prohibited uses.

Output leakage is also a significant risk. A model may include more detail than needed or reveal rare combinations that point to an individual or small group. Guardrails include output templates, suppression thresholds, review workflows, redaction, access controls, and limits on copying or exporting results.

Privacy risk can also arise from vendor relationships. Vendors may request sample data for testing or model tuning. Agencies should use synthetic, masked, or minimized data whenever possible and should require contractual controls for storage, reuse, subcontractors, breach notification, audit rights, and deletion.

## Application to AI-Supported Workflows

For generative AI, privacy-preserving design means controlling what data can be placed in prompts, what source documents are available to retrieval systems, how outputs are reviewed, and whether logs contain sensitive content. Prompt templates should make prohibited content clear and should encourage use of de-identified or aggregated inputs when possible.

For predictive analytics, privacy concerns include which features are used, whether sensitive attributes are directly included or indirectly inferred, and whether outputs could stigmatize individuals or communities. Model evaluation should include privacy review as well as accuracy and equity review.

For agentic workflows, privacy risk increases because the system may retrieve data, call tools, send messages, or update records. Privacy guardrails should include least-privilege permissions, approval checkpoints, output restrictions, audit logs, and the ability to suspend the workflow quickly if a concern arises.

## Reflection Questions

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Which AI use cases in your agency require identifiable or sensitive data, and which could be completed with de-identified or aggregated data?

What data elements create the greatest re-identification risk in your context?

How are free-text fields reviewed before they are used in AI-supported workflows?

Which outputs could reveal sensitive information even if the input data are controlled?

Who has authority to approve privacy-preserving AI controls for a workflow?

## Practical Exercise

Create a privacy-preserving AI review worksheet for one proposed public health AI workflow. The worksheet should identify the data involved, the sensitivity level, the minimum necessary fields, the approved users, the AI task, the output type, and the privacy controls needed before implementation.

The exercise should also identify residual risks and questions that need privacy, legal, security, or governance review. Learners should pay special attention to free text, small-area data, rare conditions, linked datasets, vendor access, and output sharing.

## Expected Artifact or Evidence

The expected artifact is a privacy-preserving AI review worksheet with a brief summary of the proposed workflow, data-use tier, approved data elements, prohibited data elements, access controls, output controls, audit requirements, and unresolved review questions.

## Expected Assignment

- The expected artifact is a privacy-preserving AI review worksheet with a brief summary of the proposed workflow, data-use tier, approved data elements, prohibited data elements, access controls, output controls, audit requirements, and unresolved review questions.

## Knowledge Check

- 1. Which statement best describes de-identification in public health AI?
- 2. Why can AI increase re-identification risk?
- 3. What is a strong privacy guardrail for an AI workflow?
- 4. When is de-identification alone insufficient?
- 5. What is strong completion evidence for this module?

## References and Resources

- NIST Privacy Framework
- NIST AI Risk Management Framework: <https://www.nist.gov/itl/ai-risk-management-framework>
- HHS HIPAA de-identification guidance
- CDC data governance and public health data modernization resources
- Agency privacy, legal, security, and records management policies